RAJALAKSHMI ENGINEERING COLLEGE

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V2

Started on	Wednesday, 16 September 2026, 8:33 AM
State	Finished
Completed on	Wednesday, 16 September 2026, 8:38 AM
Time taken	4 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int func(int arr[], int n, int x){
    int f = 0, l = n-1;
    int val =-1;
    while(f<=l){
        int mid = f+(l-f)/2;
        if(arr[mid]==x)
            return arr[mid];
        else if(arr[mid]<x){
            val = arr[mid];
            f = mid+1;
        }
        else
            l=mid-1;
    }
    return val;
}

int main(){
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i=0; i<n; i++)
        scanf("%d", &arr[i]);
    int x;
    scanf("%d", &x);
    printf("%d",func(arr, n, x));
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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